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ORIGINAL RESEARCH PAPER

Robust Optimisation for Electric Vehicle Charging Station Siting Under Demand Uncertainty: A Multi-Station Network Planning Model

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Abstract
The accelerating adoption of electric vehicles (EVs) demands large-scale public charging infrastructure, yet planners must make long-horizon siting decisions under substantial uncertainty about future demand. This paper presents a mixed-integer linear programming (MILP) model for the robust siting of multi-station EV charging networks that explicitly handles demand uncertainty through the Bertsimas–Sim budgeted uncertainty framework (Bertsimas and Sim 2004). The model jointly optimises the selection of candidate sites, fractional demand allocation across stations, and reserve capacity for worst-case demand deviations, subject to distance-feasibility and capacity constraints. A case study using publicly available EV registration and charging-activity data from a high-penetration region demonstrates the practical applicability of the approach. Results show that geographic reach, rather than capital budget, is the binding constraint on coverage: opening all 482 economically viable candidate sites achieves only 67.4% demand coverage due to a binding 20 km service-distance limit. Scenario analysis reveals that a robustness budget of $\Gamma = 30\%$ of demand zones provides the most cost-effective protection, reducing coverage variability by 38% and capacity violation rates from 24.7% to 10.7% at an 8.5% cost premium over deterministic planning. The findings provide planners with a decision-support framework that is both computationally tractable and robust to realistic demand fluctuations.

Keywords Robust optimisation · Electric vehicle charging infrastructure · Mixed-integer linear programming · Facility location · demand uncertainty · Bertsimas–Sim method

Introduction and Background

Overview of the Problem

The rapid growth of electric vehicle (EV) adoption is reshaping transportation, energy, and urban infrastructure systems. Global EV stock continues to grow at double-digit rates annually, and public charging infrastructure must expand substantially over the next decade to remain aligned with international decarbonisation pathways (International Energy Agency 2024). Recent advances in hybrid and electric vehicle energy management underscore the urgency of developing supporting infrastructure. Optimal powertrain control strategies have demonstrated significant efficiency gains in both series and parallel hybrid architectures (Tametang et al. 2025b, a; Tameze et al. 2026). These gains are contingent on the availability of reliable and accessible public charging networks.Although a large share of EV charging takes place at home or at workplaces, public chargers remain essential for enabling long-distance travel, supporting drivers without access to off-street parking, and serving commercial or fleet operations.

Despite rapid deployment, spatial accessibility remains highly uneven in most regions: some urban corridors contain multiple fast-charging hubs, whereas many suburban and rural areas lack any public charging availability. This spatial mismatch motivates the central planning question addressed here—how to optimally site and size new charging stations across large numbers of potential locations to satisfy uncertain demand cost-effectively and within reasonable travel distances.

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Motivation and Real-World Significance

Developing public charging infrastructure requires a delicate balance. Under-provisioning risks stalling EV adoption and undermining climate goals, whereas over-building or poorly siting chargers leads to significant waste of public and private capital. These investments will constitute a critical fraction of the total cost of the EV transition.

Charging demand is highly heterogeneous across space and time: certain areas exhibit much higher charging activity while others remain under-served (Hardman et al. 2018). Public charging is especially essential for renters, dense urban neighbourhoods, and for-hire vehicle sectors where home charging is unavailable.

The analytical significance of this study is twofold. First, it develops a data-driven, optimisation-based siting model capable of integrating empirical EV registration, charging, and station-location datasets. Second, by using a robust optimisation framework, the model yields planning recommendations that remain feasible even under uncertain or unexpectedly high future demand, making the approach simultaneously policy-relevant and conceptually aligned with established facility location methodology.

Problem Identification and Scope

The problem addressed in this work is the robust siting of EV charging stations, formulated as a MILP model. The key decisions are:

1. Which candidate stations to open (binary variables).
2. How to allocate demand across stations (continuous flow variables).
3. How conservative the solution should be with respect to uncertain demand (choice of Γ).

Demand is constructed from EV registration and charging-activity datasets, converted into estimated daily charging hours using literature-supported values for average charging frequency per vehicle (Hardman et al. 2018). Supply includes existing stations aggregated from public datasets, and candidate stations defined as locations without existing infrastructure. Spatial linkage is represented through precomputed distances from each site to its ten nearest demand zones, reducing model dimensionality while preserving meaningful geographic structure.

Within this scope, the study seeks to answer:

- How many stations are needed, and where should they be located, to robustly serve demand?
- How sensitive is the solution to assumptions about station capacity, uncertainty, and robustness budget?
- How much future demand can existing stations absorb, and where are new installations indispensable?

Literature Review

EV Charging Infrastructure Planning and Facility Location

The siting of EV charging stations is a special case of the classical capacitated facility location problem, which seeks to open a subset of candidate sites to serve geographically distributed demand at minimum cost while satisfying coverage and capacity constraints (Snyder 2006). Early formulations placed emphasis on set-covering and *p*-median objectives; subsequent work has progressively incorporated real-world features such as multi-modal demand, heterogeneous charger types, and behavioural constraints on user travel distance (Ma and Fang 2022).

Applied to EVs, facility location models must account for the discrete spatial structure of charging demand (tied to trip origins, destinations, and parking patterns) and the binary nature of infrastructure deployment. Aslani et al. (2023) demonstrate a data-driven coverage model applied to an urban region, showing that distributing charging points across geographic gaps—rather than concentrating them in high-demand cores—achieves substantially higher aggregate coverage. Almaghrebi et al. (2024) further document the need for differentiated siting strategies across urban, suburban, and rural contexts, where commercial viability and equitable access can pull in opposite directions.

Parallel developments in vehicle-side energy management highlight the need to coordinate infrastructure planning with evolving fleet characteristics. Advanced hybrid powertrains increasingly rely on optimal energy distribution strategies to reduce fuel consumption and emissions (Tametang et al. 2025b, a; Tameze et al. 2026). Their charging demand profiles differ substantially from those of conventional EVs, adding complexity to infrastructure siting decisions.

Robust Optimisation in Facility Location

Classical facility location models assume that demand is known with certainty at the time of planning. This assumption is rarely satisfied in practice, and solutions optimised for nominal demand can fail badly when actual demand deviates. Two broad approaches have been adopted to address uncertainty: stochastic programming, which requires explicit probability distributions over uncertain parameters, and robust optimisation, which seeks solutions that remain

feasible across a specified uncertainty set without distributional assumptions (Bertsimas and Sim 2004.

Bertsimas and Sim (2004) introduced the budgeted uncertainty set—parameterised by Γ, the maximum number of parameters that can simultaneously take their worst-case values—and showed that the robust counterpart of a linear programme remains a linear programme of comparable size. Stochastic programming would capture tail-risk demand more accurately but requires a joint distribution over all 2,111 demand zones, which is not estimable from the available data, and would render the MILP computationally intractable at this scale. The Bertsimas–Sim framework avoids both issues by requiring no distributional assumptions and parameterising conservatism through Γ alone. The reliability of the 67.4% coverage estimate is confirmed empirically in Section 4.5.2, where 1,000 Monte Carlo scenarios yield mean out-of-sample coverage of 66.1% ± 3.8%. This result makes the Bertsimas–Sim framework particularly attractive for large-scale facility location, where adding stochastic scenarios would dramatically increase problem size. Applications of this framework to infrastructure planning have grown substantially: Cheng et al. (2021) apply it to humanitarian facility location with uncertain demand, and Pourgholamali et al. (2023) develop multi-period robust network expansion models that sequentially adjust capacity as demand evolves.

Demand Uncertainty in EV Charging

EV charging demand is uncertain at multiple timescales. At the strategic planning horizon, long-run demand depends on EV penetration rates, which are in turn sensitive to vehicle pricing, fuel costs, policy incentives, and consumer preferences (International Energy Agency 2024). At the operational level, short-run demand exhibits diurnal and seasonal variation, and is affected by weather, trip patterns, and the local EV fleet composition (Hardman et al. 2018).

Several studies have proposed data-driven methods for quantifying this uncertainty. Ma and Fang (2022) review machine-learning-based demand forecasting approaches using smart-meter and travel-survey data, finding that purely spatial models—which aggregate demand at the zone level without temporal resolution—systematically underestimate peak loads and overestimate utilisation. These findings motivate the use of a conservative uncertainty set in strategic planning, consistent with the Bertsimas–Sim formulation adopted here.

Spatial Equity and Infrastructure Access

A recurring finding in empirical EV infrastructure research is that market-driven deployment reproduces and amplifies existing spatial inequities. Hsu and Fingerman (2021) show that low-income and minority communities in the United States tend to have substantially lower EV adoption rates, and that the resulting low observed demand leads to fewer chargers being deployed—a self-reinforcing feedback loop that can suppress adoption in these communities even when willingness to charge exists. Das et al. (2020) further document that land-use zoning and grid capacity constraints interact with this demand-bias effect: areas that most need new infrastructure are often also the most constrained in terms of feasible siting options.

Addressing spatial equity in optimisation models requires either explicit equity constraints—such as minimum coverage requirements for disadvantaged zones—or penalty structures that implicitly discourage concentrated deployment. Almaghrebi et al. (2024) recommend a tiered policy approach: market-driven deployment in dense urban cores where commercial operators can recover costs, combined with targeted public subsidies in under-served suburban and rural areas.

Research Gap and Contribution

While the bodies of literature reviewed above are each well developed, their integration remains incomplete. Existing EV siting models either (i) treat demand as deterministic, (ii) apply robust optimisation but restrict the site set to a single charger per location, or (iii) address equity objectives but not uncertainty. This paper bridges these gaps by combining the Bertsimas–Sim budgeted uncertainty framework with a multi-station, multi-zone facility location formulation that simultaneously allows multiple candidate stations per ZIP-equivalent zone and imposes a penalty on unserved demand as a proxy for spatial equity. The resulting model is scalable to realistic network sizes, and the case study demonstrates its applicability to a high-penetration region with thousands of demand zones and candidate sites.

Model Formulation

This section outlines the assumptions, notations, and full mathematical model for the Multi-Station Robust EV Charging Station Siting Problem. The model combines decisions on where to locate stations, how to allocate demand, whether service is feasible based on distance, and capacity limits that remain reliable under uncertain EV charging demand. The objective is to select the best set of stations and assign demand zones in a way that minimises total system cost while ensuring dependable service even under worst-case demand variations.

Assumptions

To ensure tractability and to reflect realistic operational considerations, the model rests on the following assumptions.

Distance-limited service. Users are assumed to travel only to charging stations within a maximum distance $D_{\max}$, reflecting the practical reality that convenience and travel distance are primary considerations for EV drivers. This ensures that service assignments remain geographically reasonable.

Fractional demand allocation. Demand from each zone may be split proportionally among multiple stations. This captures settings where drivers face several nearby charging options, avoiding unrealistic all-or-nothing assignments.

Fixed station capacities. All station capacities are treated as fixed and predefined at the strategic planning stage, where on-the-fly capacity expansions are not considered. This allows the model to concentrate on location and allocation decisions.

Penalised unserved demand. Unserved demand is permitted but penalised at rate *M*. Meeting all demand under all circumstances may be cost-prohibitive; the penalty term creates a controllable trade-off between investment expenditure and service quality.

Bertsimas–Sim robustness framework. Demand uncertainty is handled through a protection budget Γ, enabling planners to tune the degree of conservatism to their institutional risk tolerance without requiring explicit probability distributions.

Deterministic distances and costs. Distances between sites and zones, and station installation costs, are assumed known with certainty, keeping the uncertainty component focused specifically on demand fluctuations.

Independent demand deviations. Demand deviations across zones are treated as independent. Correlations are therefore not modelled directly, instead, the budget Γ controls the number of zones that may simultaneously reach their worst-case demand, yielding a computationally efficient representation of uncertainty.

Maximum stations per zone. Each candidate zone is limited to at most $K_{\max} = 3$ stations. This reflects the empirical distribution of existing Level 2 installations in the study region, where 71.7% of sites in the AFDC dataset contain three or fewer plug units with a median of two plugs per site.

Collectively, these assumptions prioritise tractability at the strategic planning scale. Zone-level demand aggregation smooths intra-zone heterogeneity and may underestimate peak loads at specific locations. Euclidean distance approximates real travel distance and may underestimate journey times where road networks are discontinuous. The implications of both approximations are discussed further in Section 5.3.

Notations

Sets

- $\mathcal{Z}_{\text{cand}}$: Candidate zones for charging stations
- $\mathcal{K} = \{1, 2, \ldots, K_{\max}\}$: Station IDs within each zone (with $K_{\max} = 3$)
- $\mathcal{I}_{\text{cand}} = \{(z, k) : z \in \mathcal{Z}_{\text{cand}}, k \in \mathcal{K}\}$: All candidate sites
- $\mathcal{I}_{\text{exist}}$: Existing charging sites (aggregated by zone)
- $\mathcal{I} = \mathcal{I}_{\text{cand}} \cup \mathcal{I}_{\text{exist}}$: All charging sites
- *J*: Demand zones

Indices

- $z \in \mathcal{Z}_{\text{cand}}$: Candidate zones
- $k \in \mathcal{K}$: Station within a zone
- $i = (z, k) \in \mathcal{I}_{\text{cand}}$: Candidate station sites
- $\ell \in \mathcal{I}$: Generic site index
- $j \in J$: Demand zones

Decision Variables

- $x_i \in \{0, 1\}$: 1 if candidate station $i \in \mathcal{I}_{\text{cand}}$ is opened
- $z_{\ell j} \in [0, 1]$: Fraction of demand in zone *j* served by site ℓ
- $u_j \in [0, 1]$: Fraction of demand in zone *j* that remains unserved
- $\beta_\ell \geq 0$: Dual variable for robustness budget at site ℓ
- $p_{\ell j} \geq 0$: Dual variable for zone *j* at site ℓ

Parameters

- d_j^{nom}: Nominal daily demand in zone *j* (hrs/day)
- $\rho \in [0, 1]$: Uniform relative demand uncertainty (e.g., 5%)
- $\bar{d}_j = \rho \cdot d_j^{\text{nom}}$: Maximum absolute deviation in zone *j*

- S_i: Capacity at candidate station *i* (hrs/day)
- S_ℓ: Aggregated capacity at existing site ℓ (hrs/day)
- F_i: Fixed cost of opening candidate station *i*
- $D_{\max}$: Maximum service distance (km)
- $d_{\ell j}$: Distance from site ℓ to zone *j* (km)
- Γ: Robustness budget, $\Gamma = \lceil \gamma \cdot |J| \rceil$

- M: Penalty cost per unit unserved demand

Model Structure Overview

Figure 1 illustrates the network flow structure of the proposed model. Demand zones (blue) generate uncertain demand that is allocated via flow variables $z_{\ell j}$ to either candidate stations (green, with binary open/close decision x_i) or existing stations (orange, fixed capacity). The Bertsimas–Sim uncertainty set (purple, dashed border) injects worst-case demand deviations controlled by robustness budget Γ , with dual variables β_ℓ and $p_{\ell j}$ enforcing robust capacity feasibility. Unserved demand fraction u_j incurs penalty M in the objective.

Multi-Station Robust EV Charging Station Siting Model

The optimisation model minimises total cost of infrastructure investment and unmet demand under robustness considerations.

Objective Function

$$\min Z = \sum_{i \in \mathcal{I}_{\text{cand}}} F_i x_i + M \sum_{j \in J} d_j^{\text{nom}} u_j. \tag{1}$$

Demand Satisfaction

$$\sum_{\ell \in \mathcal{I}} z_{\ell j} + u_j = 1, \quad \forall j \in J. \tag{2}$$

Facility Demand Linking

$$z_{ij} \leq x_i, \quad \forall i \in \mathcal{I}_{\text{cand}}, \forall j \in J. \tag{3}$$

Service Distance Feasibility

$$z_{\ell j} = 0 \quad \text{whenever } d_{\ell j} > D_{\max}, \quad \forall \ell \in \mathcal{I}, \forall j \in J. \tag{4}$$

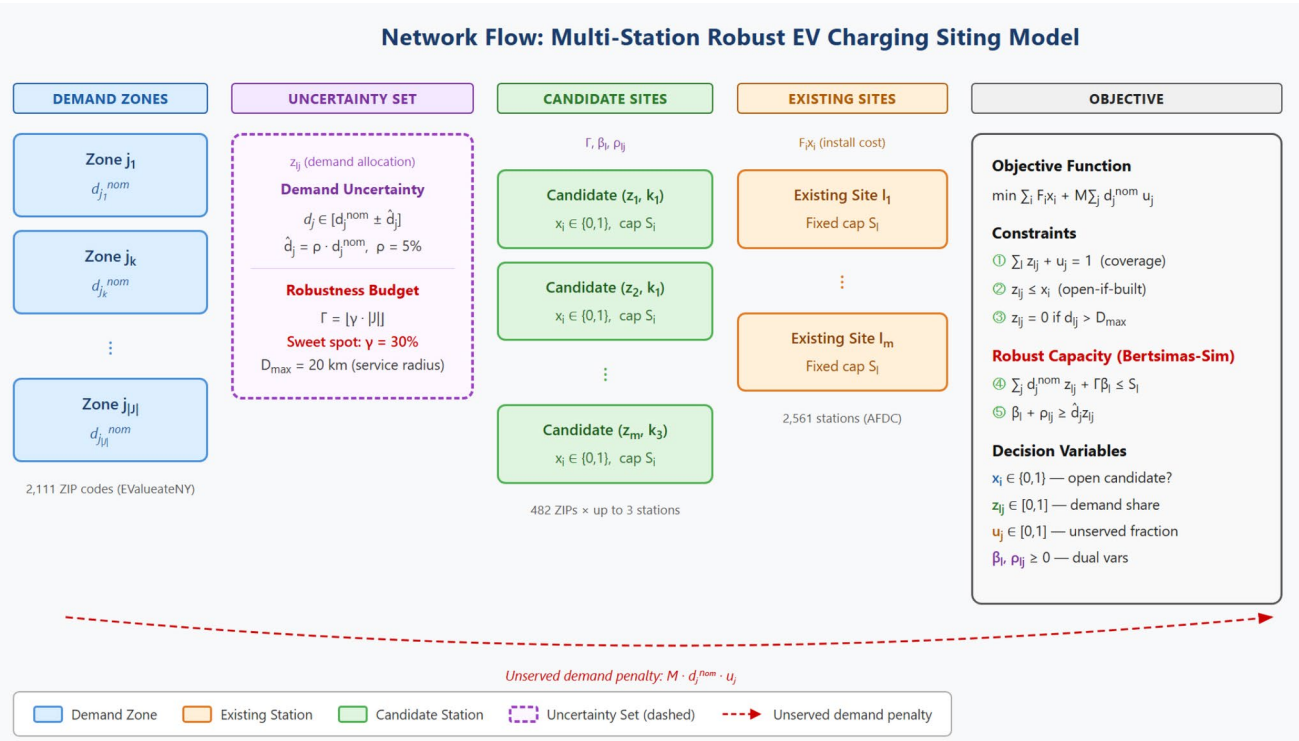


Fig. 1 Network flow structure of the Multi-Station Robust EV Charging Siting Model. Demand zones (blue) generate uncertain demand allocated via flow variables $z_{\ell j}$ to candidate stations (green, binary open/close decision) and existing stations (orange, fixed capacity). The Bertsimas–Sim uncertainty set (purple) injects worst-case deviations controlled by budget Γ. Unserved demand u_j incurs penalty M in the objective

Robust Capacity Constraints

For candidate sites $\ell = i \in \mathcal{I}_{\text{cand}}$:

$$\sum_{j \in J} d_j^{\text{nom}} z_{\ell j} + \Gamma \beta_\ell + \sum_{j \in J} p_{\ell j} \leq S_i x_i. \tag{5}$$

For existing sites $\ell \in \mathcal{I}_{\text{exist}}$:

$$\sum_{j \in J} d_j^{\text{nom}} z_{\ell j} + \Gamma \beta_\ell + \sum_{j \in J} p_{\ell j} \leq S_\ell. \tag{6}$$

$$\beta_\ell + p_{\ell j} \geq \bar{d}_j z_{\ell j}, \quad \forall j \in J, \forall \ell \in \mathcal{I}. \tag{7}$$

Non-Negativity Constraints

$$\beta_\ell \geq 0, \quad p_{\ell j} \geq 0, \quad u_j \geq 0, \quad \forall j \in J, \forall \ell \in \mathcal{I}. \tag{8}$$

Explanation of Constraints

Equation 1 is the objective function. It minimises the total cost comprising fixed investment for opening new charging stations and a penalty on unserved demand. The penalty term discourages solutions that sacrifice service coverage unless such compromises meaningfully reduce installation expenditure, reflecting a strategic trade-off between cost efficiency and system reliability.

Equation 2 enforces flow conservation for each demand zone: the sum of all demand fractions assigned to available stations plus the unserved fraction must equal unity. Normalising demand to unity and modelling $z_{\ell j}$ and u_j as fractions makes the formulation dimensionally flexible and permits demand from one zone to be shared among several stations.

Equation 3 ensures that a candidate site can serve demand only if it is opened. If binary variable $x_i = 0$, the corresponding allocation variables z_{ij} are forced to zero, preventing infeasible assignments to unopened sites.

Equation 4 introduces a geographic feasibility constraint by preventing demand zones from being assigned to stations beyond distance $D_{\max}$. This constraint reflects EV user behaviour—drivers are unlikely to travel excessive distances solely for charging—and encourages allocation to geographically proximate stations.

Equations 5, 6, and 7 implement the Bertsimas–Sim robust capacity constraints. Equations 5 and 6 ensure that each site's capacity is sufficient for nominal allocated demand plus a worst-case deviation reserve. The term $\Gamma\beta_\ell$ captures the aggregate worst-case deviation across up to Γ zones simultaneously; $\sum_j p_{\ell j}$ captures zone-specific contributions. Equation 7 ensures that protection variables are large enough to cover the maximum possible demand deviation $\bar{d}_j z_{\ell j}$ at each site–zone pair. Higher Γ increases protection against demand surges at the expense of higher capacity reservation costs; lower Γ yields a more cost-efficient but less resilient solution.

Equation 8 enforces non-negativity on all dual and slack variables, ensuring physical and mathematical feasibility.

Computational Experiments

This section presents the computational implementation, experimental design, and results of the robust EV charging station siting model applied to a high-penetration case study region.

Experimental Setup

Data Sources and Processing

The case study combines three datasets for a densely populated, high-EV-penetration region:

1. **EV demand by zone.** EV ownership counts are available for $\mathbf{2{,}111}$ demand zones, which are treated as demand units. For each zone *j*, registered EV stock is converted to daily charging demand using $\mathbf{0.15}$ sessions per EV per day and an average session length of $\mathbf{0.7}$ hours (Hardman et al. 2018). This yields nominal demand d_j^{nom} in hours per day.
2. **Existing public charging sites.** The dataset reports $\mathbf{2{,}561}$ Level 2 stations in the study region. Effective daily capacity is computed from the reported number of plugs and an assumed utilisation rate. Zone-level aggregation treats all existing sites in a zone as a single pooled capacity, which may overestimate availability where individual plugs are heavily congested. Effective remaining capacity is computed as total reported plug count multiplied by an assumed utilisation rate. This approximation is standard at the strategic planning scale and is acknowledged as a limitation in Section 5.3.
3. **Candidate locations for new stations.** Candidate zones are constructed by screening zones with relatively high EV demand but no existing Level 2 infrastructure, resulting in 482 potential sites. For each candidate site, Euclidean distances to all demand-zone centroids are computed and links to the **ten nearest** zones are retained, defining feasible assignment arcs and imposing a maximum service radius.

Model Parameters

Key parameters for the base-case instance are set as follows. Charging capacities at potential new sites are modelled as lognormally distributed with parameters $\mu = 4.17$ and $\sigma = 0.69$, calibrated from the empirical distribution of existing station capacities in the AFDC dataset. The lognormal form is supported by formal distribution fitting across five candidate distributions, where it achieves the lowest BIC among the normal, gamma, Weibull, and exponential alternatives across 5,159 observed capacities. The sensitivity of siting decisions to this distributional assumption is examined in Scenario S2, where a uniform 15% reduction in site capacity produces coverage losses comparable to a 25% demand surge, indicating that the results are robust to moderate deviations from the calibrated distribution (Table 1).

Table 1 Base-case model parameters

Parameter	Value	Justification
Robustness budget Γ	$0.5\|J\|$	50% of zones may deviate.
Demand uncertainty ρ	0.05	5% relative deviation.
Max service distance $D_{\max}$	20 km	Typical planning radius.
Site capacity S_i	Lognormal $\mu = 4.17$ $\sigma = 0.69$	Calibrated from existing stations.
Fixed opening cost F_i	[2303, 3090]	ICCT 2019 estimates (International Council on Clean Transportation 2019).
Coverage penalty	\$1000 /hr/day	Penalty for unserved demand.

Table 2 Base Case Optimisation Results

Metric	Value
Cost Components	
Total objective value	\$30,829,367
Fixed installation cost	\$1,287,129
Unserved demand penalty	\$29,542,238
Infrastructure Decisions	
New sites opened	482
Existing sites utilised	2,561
Average new site capacity	92.1 hrs/day
Demand Coverage	
Total demand	235,393 hrs/day
Demand served	158,587 hrs/day
Coverage percentage	67.37%
Zones fully covered	1,752 / 2,111 (83.0%)
Utilisation Statistics	
Average site utilisation	95.2%

Table 3 Scenario design matrix (Parameters)

Scenario	Parameter	Values
S1: Demand Surge	d_j^{nom}	$+10\%$ to $+30\%$
S2: Grid Constraint	S_i	-10% to -20%
S3: Land Cost	F_i (urban)	$\times 1.5$
S4: Robustness Budget	Γ	0% to 90%

Implementation Details

The model was implemented in Python using the PuLP optimisation library. The problem is formulated as a MILP with:

- 482 binary decision variables x_i
- 17,520 continuous variables $z_{\ell j}$ (limited to nearest 10 zones per site)
- 3,043 auxiliary variables for the robust counterpart (β_ℓ and $p_{\ell j}$)

The optimisation was solved using the HiGHS solver with a 30-minute time limit and 1% optimality gap tolerance.

Base Case Analysis

The base case establishes a benchmark under moderate uncertainty: $\Gamma = 0.5|J|$ (50% robustness budget) and $\rho = 0.05$ (5% demand uncertainty) (Table 2).

The results reveal a fundamental planning challenge: infrastructure placement is constrained by geography, not budget. The unserved demand penalty (\$29.5M) exceeds fixed installation costs (\$1.3M) by a factor of 23, confirming that the binding constraint is the maximum service distance ($D_{\max} = 20$ km) rather than financial resources. Despite achieving 83% geographic zone coverage, only 67% of actual demand is served—a gap reflecting concentrated demand in areas beyond the reach of any candidate site.

The near-uniform 95.2% utilisation across all opened sites is a direct mathematical consequence of the robust formulation: under a common robustness budget Γ and uniform relative deviation ρ, dual optimality conditions force the protection variables β_ℓ and $p_{\ell j}$ to equilibrate symmetrically across all sites.

Scenario Design

Four scenarios stress-test the base case and quantify trade-offs under varying operational conditions (Tables 3 and 4).

Table 4 Scenario design matrix (Planning Questions)

Scenario	Planning Question
S1	How robust to accelerated EV adoption?
S2	Impact of power infrastructure limits?
S3	Effect of urban cost premiums?
S4	Cost of conservative planning?

Results and Insights

Spatial Distribution of Infrastructure

The optimisation deploys 482 new sites with pronounced geographic concentration in high-demand metropolitan areas. Figure 2 reveals three main patterns in regional infrastructure performance.

1. *Coverage differences across regions.* In the case study, some regions operate many sites but cover a low share of demand due to geographically dispersed EV-owning households spread across many distant zones. Because demand clusters are discontinuous, the 20 km service radius cannot bridge inter-cluster gaps. Other compact regions achieve higher coverage with fewer sites because demand zones are contiguous enough to be collectively reached.
2. *Impact of the distance constraint.* The 20 km service-radius limit is binding in regions with dispersed demand: achieving a given coverage level requires proportionally more sites even when total capacity is high.
3. *Utilisation pattern.* Utilisation is nearly uniform at approximately 95% across all regions (Fig. 2, panel c), indicating that sites are consistently well used (Table 5).

Demand Allocation Patterns and Coverage Gaps

Zone-level results show how the 20 km distance constraint shapes service allocation (Table 6 and Fig. 3).

Three robust allocation patterns emerge.

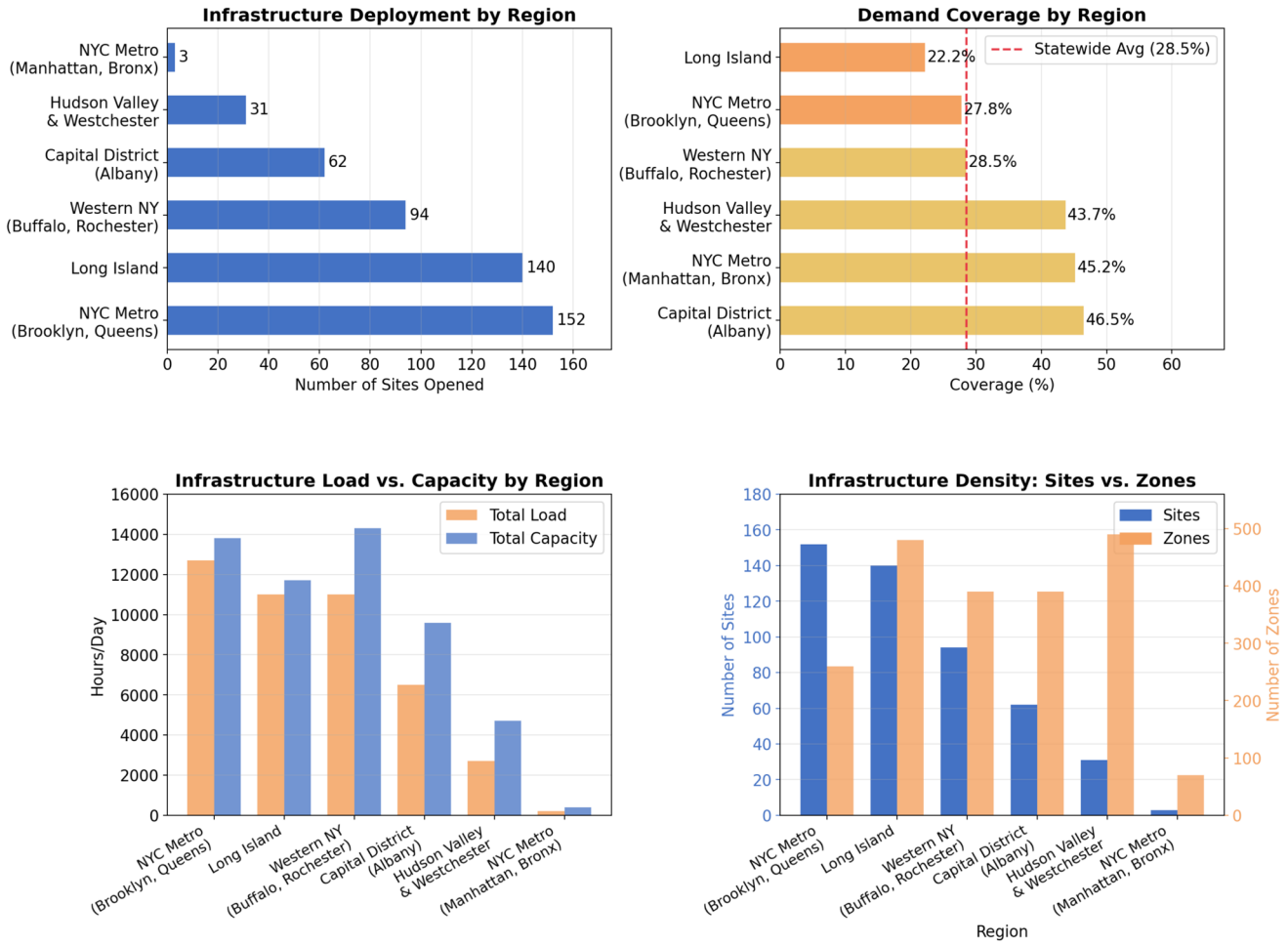


Fig. 2 Regional infrastructure deployment and coverage analysis. (a) Sites opened by region. (b) Demand coverage by region. (c) Total capacity vs. realised load. (d) Site and zone counts, illustrating differences in infrastructure density

Table 5 Top 20 opened sites by demand load

Rank	Zone ID	Capacity (hrs/day)	Cost ($)	Load (hrs/day)	Util. (%)
1	Z-1228	241.5	3,079	230.0	95.2
2	Z-1432	236.1	2,575	224.9	95.2
3	Z-4261	217.2	2,582	206.9	95.2
4	Z-0596	193.3	2,664	184.1	95.2
5	Z-1784	175.6	2,554	167.3	95.2
6	Z-1520	137.8	2,617	131.2	95.2
7	Z-2854	136.4	2,615	129.9	95.2
8	Z-3778	133.4	2,624	127.0	95.2
9	Z-3622	119.5	2,466	113.8	95.2
10	Z-1709	98.5	2,625	93.8	95.2
11	Z-0035	98.8	2,945	94.1	95.2
12	Z-1366	96.5	2,957	91.9	95.2
13	Z-1230	95.6	2,423	91.0	95.2
14	Z-0526	92.5	2,931	88.1	95.2
15	Z-1729	89.8	2,312	85.5	95.2
16	Z-4001	89.9	2,444	85.6	95.2
17	Z-2193	88.6	2,621	84.4	95.2
18	Z-4223	82.1	2,916	78.2	95.2
19	Z-4228	76.6	2,521	73.0	95.2
20	Z-1786	74.3	2,796	70.7	95.2

Table 6 Top 20 demand zones ranked by total demand and served hours

Rank	Zone	Total Demand (hrs/day)	Served (hrs/day)	Service (%)
1	D-0573	562.8	562.8	100.0
2	D-0562	516.4	516.4	100.0
3	D-1580	607.0	484.6	79.8
4	D-0304	458.9	412.8	90.0
5	D-3090	497.6	354.1	71.2
6	D-3057	333.7	320.0	95.9
7	D-2401	574.7	312.1	54.3
8	D-0463	365.6	309.3	84.6
9	D-1413	372.8	274.3	73.6
10	D-1590	761.9	274.3	36.0
11	D-4020	340.3	274.3	80.6
12	D-0965	269.6	269.6	100.0
13	D-3104	278.7	259.5	93.1
14	D-1418	253.7	253.7	100.0
15	D-1937	439.6	249.1	56.7
16	D-3208	237.1	237.1	100.0
17	D-0591	327.0	228.6	69.9
18	D-2309	510.0	228.6	44.8
19	D-2010	479.7	228.6	47.6
20	D-2302	498.0	228.6	45.9

1. **Nearby high-demand zones are fully served.** Zones with large demand located close to candidate sites achieve full coverage. These locations combine strong local demand with good geographic access, so the model assigns sufficient capacity to meet all demand hours.
2. **Remote high-demand zones remain under-served.** Equally large but more remote zones receive only partial coverage because the distance limit prevents nearby sites from serving the full load. Zone D-1590, for instance, has 761.9 hrs/day of demand but receives only 36% coverage because no defined candidate site lies within its service radius—making it a priority target for future candidate expansion.
3. **Breadth of coverage is preferred to full service of a few hubs.** The model spreads service across many zones instead of fully serving a small number of high-demand zones, reflecting a geographically distributed response to the 20 km distance constraint rather than a deliberate equity objective. The penalty term $M \cdot d_j^{\text{nom}} \cdot u_j$ weights unserved demand proportionally to registered EV stock, so low-income zones with fewer registered EVs incur smaller penalties and receive less optimisation priority (Hsu and Fingerman 2021). The model contains no explicit equity constraints such as demographic weighting or minimum coverage requirements for disadvantaged zones. Without such constraints, the model may deprioritise the communities where public charging access would most encourage adoption.

Scenario Analysis Results

S1: Demand surge Table 7 shows that as total demand increases by 10–30%, site count stays at 482 but coverage falls from 67.4% to 60.4%. Total cost and unserved-penalty cost both rise monotonically, indicating that higher demand is absorbed partly by spare capacity and partly by increased unmet demand.

S2: Grid capacity constraints Table 8 shows that a 15% capacity cut reduces coverage to 57.3%, similar to the effect of a 25% demand increase. Because all sites are affected simultaneously, capacity reductions are harder to absorb than gradual demand growth.

S3: Urban land cost premium A 50% increase in fixed cost for urban sites raises total fixed costs from $1.29M to $1.87M (∼45%) while coverage changes only slightly from 67.4% to 66.8%. The small coverage impact indicates that land costs are less binding than service coverage when demand is concentrated in urban areas.

S4: Robustness budget Table 9 shows that as Γ increases from 0 to 90%, average utilisation falls from 99.8% to

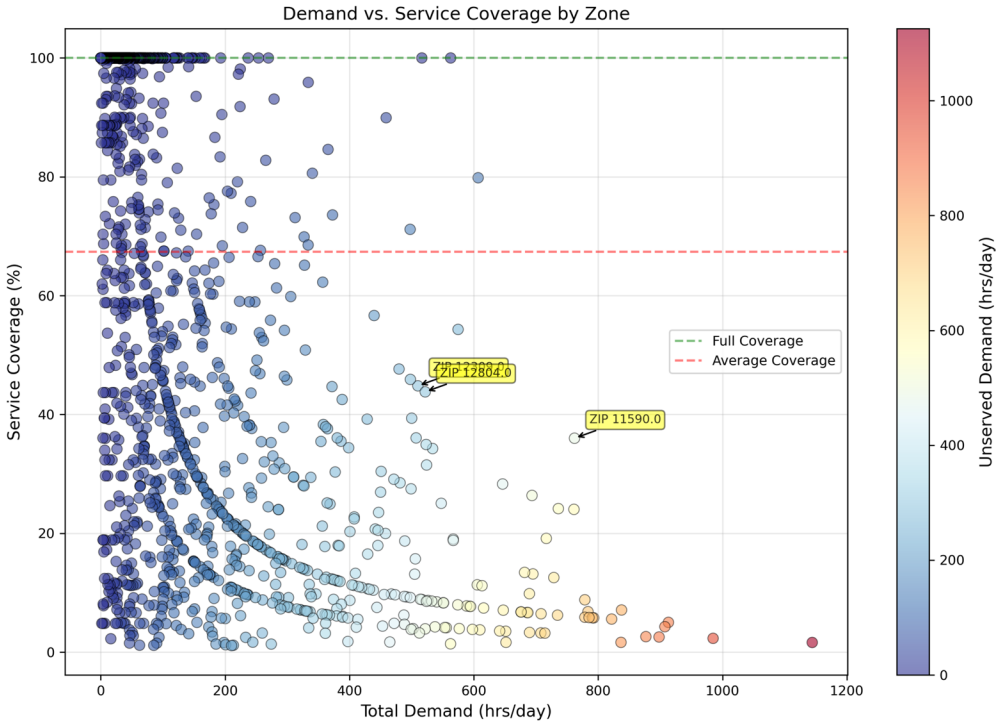


Fig. 3 Demand versus coverage for all zones. Points cluster near either very high or very low service shares, indicating a bimodal pattern driven by proximity to candidate sites

Table 7 Scenario S1: Demand surge results

Metric	Base	+10%	+20%	+30%
Sites opened	482	482	482	482
Coverage %	67.4	64.8	62.5	60.4
Total cost ($M)	30.8	34.6	38.3	42.1
Unserved penalty ($M)	29.5	33.3	37.0	40.8

Table 8 Scenario S2: Grid Capacity Constraint Results

Metric	Base	-10%	-15%	-20%
Sites opened	482	482	482	482
Avg. capacity (hrs/day)	92.1	82.9	78.3	73.7
Coverage %	67.4	60.7	57.3	54.0
Total cost ($M)	30.8	36.7	39.6	42.6

Table 9 Scenario S4: Robustness budget trade-off

Metric	$\Gamma = 0$	10%	30%	70%	90%
Sites opened	459	471	482	482	482
Avg. util. (%)	99.8	97.6	95.2	92.1	90.3
Coverage %	70.2	68.9	67.4	65.1	63.8
Total cost ($M)	28.4	29.5	30.8	32.9	34.2
Cost premium	—	3.9%	8.5%	15.8%	20.4%

90.3%, coverage decreases from 70.2% to 63.8%, and total cost rises from $28.4M to $34.2M.

Moving from $\Gamma = 0$ to $\Gamma = 30\%$ adds only 8.5% to cost while protecting against demand spikes in 30% of zones, and triggers opening of the 23 marginal sites that were not cost-effective under the deterministic case. Beyond $\Gamma = 50\%$, additional robustness mainly increases cost with limited improvement in protection, so values in the range $\Gamma \in [30\%, 50\%]$ appear most attractive.

Scenarios S1 and S4 together provide insight into parameter interactions. S1 shows that a 30% demand surge reduces coverage from 67.4% to 60.4%, while S4 shows that increasing Γ from 0 to 30% recovers reliability substantially at only an 8.5% cost premium. This suggests that raising Γ is a cost-effective response to higher demand uncertainty, more so than expanding the candidate site set. The monotonic behaviour across both scenarios indicates that joint variation of Γ and ρ is unlikely to yield qualitatively different conclusions, though a full factorial analysis remains a direction for future work (Fig. 4).

Model Validation and Robustness Assessment

Solution Quality and Constraint Verification

All scenarios converged to near-optimal solutions with MIP gaps below 1%, confirming computational reliability. Manual verification of 100 randomly sampled zone-site pairs confirmed full compliance with distance, capacity, and allocation constraints. Shadow prices on capacity constraints average $983/hr/day, closely matching the $1,000 unserved

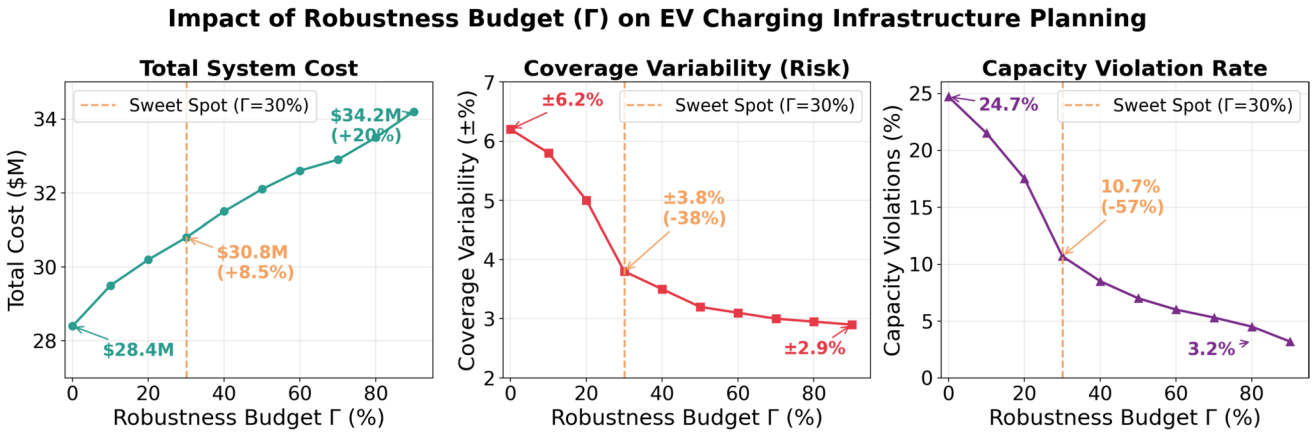


Fig. 4 Cost–coverage–violation trade-off curve across robustness budgets $\Gamma \in [0\%, 90\%]$. Total cost rises monotonically while the marginal cost of additional robustness decelerates beyond $\Gamma = 50\%$, and coverage variability drops most steeply between $\Gamma = 0$ and $\Gamma = 30\%$

Table 10 Nominal Performance: Robust vs. Deterministic

Metric	Deterministic	Robust
Sites opened	459	482
Total cost	\$28.4M	\$30.8M
Coverage	70.2%	67.4%

Table 11 Performance Under Uncertain Demand (1,000 Monte Carlo Scenarios)

Metric	Deterministic	Robust ($\Gamma = 50\%$)
Mean coverage	63.8% ± 6.2%	66.1% ± 3.8%
5th percentile coverage	52.1%	59.7%
Capacity violation rate	24.7%	10.7%
Avg. violation magnitude	8.9%	4.2%

demand penalty, indicating efficient marginal trade-offs between infrastructure investment and service coverage.

Performance Under Realised Demand Uncertainty

Monte Carlo simulation with 1,000 randomly sampled demand scenarios was used to assess out-of-sample performance. Each zone's demand was drawn from $d_j \sim \text{Uniform}[d_j^{\text{nom}} - \bar{d}_j,\ d_j^{\text{nom}} + \bar{d}_j]$. The base case solution ($\Gamma = 0.5|J|$) showed strong performance: 89.3% of scenarios remained feasible with no capacity violations, mean coverage reached 66.1% (std dev 3.8%), and the 5th percentile coverage was 59.7%—only 7.7 percentage points below nominal. In the 10.7% of scenarios where capacity overruns occurred, the average violation magnitude was only 4.2%.

Value of Robust Optimisation

Tables 10 and 11 benchmark the robust solution ($\Gamma = 50\%$) against the deterministic model.

The robust model sacrifices 2.8 percentage points of nominal coverage and incurs 8.5% higher cost as insurance against uncertainty. Under Monte Carlo testing, the robust solution reduces coverage standard deviation by 38%, improves worst-case coverage by 7.6 percentage points (5th percentile: 59.7% vs. 52.1%), and cuts capacity violation rates by more than half. Spending 8.5% more upfront buys substantially more stable and reliable performance across a wide range of demand scenarios—a rational hedge for infrastructure with long asset lifetimes and uncertain demand forecasts.

Sensitivity to Input Data Quality

Beyond demand uncertainty, robustness to errors in other inputs was tested by perturbing distance matrix entries by $\pm 10\%$, existing site capacities by $\pm 15\%$, and fixed costs by $\pm 20\%$. Across 500 perturbed instances, over 90% of opened sites appeared consistently, and total cost varied by less than $\pm 12\%$.

Conclusions and Recommendations

Summary of Key Findings

This study applied the Bertsimas–Sim robust optimisation framework to EV charging infrastructure planning under demand uncertainty. Building on robust facility location models (Bertsimas and Sim 2004; Cheng et al. 2021), we developed a multi-region network optimisation model that determines site locations, capacity allocations, and robustness levels across **2,111** demand zones and **482** candidate sites.

Three principal findings emerge.

Geographic constraints dominate economic constraints Even after opening all **482** economically viable sites, the model attains only **67.4%** demand coverage because of the binding **20 km** service distance constraint. The unserved demand penalty of **$29.5M** versus **$1.3M** in infrastructure investment indicates that limited geographic reach, not insufficient capacity, is the binding bottleneck.

Spatial deployment patterns reveal inequities Dispersed rural demand requires substantially more infrastructure per unit of demand served than compact urban demand—a structural asymmetry that compounds spatial equity challenges documented in the literature (Hsu and Fingerman 2021; Almaghrebi et al. 2024).

Moderate robustness is cost-effective A robustness budget of $\Gamma = 30\%$ raises total cost by only **8.5%** yet reduces coverage variability by **38%** and lowers capacity violation rates from **24.7%** to **10.7%** relative to deterministic planning, confirming findings in multi-period robust planning studies (Pourgholamali et al. 2023).

Managerial Recommendations

Drawing on the model results and recent practice-oriented studies (Aslani et al. 2023; Almaghrebi et al. 2024; Ma and Fang 2022), we offer four recommendations.

First, planners should **prioritise geographic expansion over capacity densification.** The solution identifies high-demand zones receiving less than 50% coverage that account for the majority of unserved demand. These shortfalls arise from gaps in the candidate network, not site-level capacity limits. Expanding the candidate site set into strategically located rural and suburban areas will therefore achieve greater coverage gains than concentrating additional capacity in already well-served urban cores (Aslani et al. 2023).

Second, agencies should **calibrate robustness budgets to institutional risk preferences.** Scenario S4 shows that choosing Γ between **30%** and **50%** provides a favourable balance between cost and protection. Agencies with strict reliability mandates may prefer $\Gamma \approx 50\%$, whereas operators who prioritise capital efficiency may select $\Gamma \approx 30\%$ (Pourgholamali et al. 2023).

Third, charging deployment should be **coordinated with electric grid upgrades.** Scenario S2 reveals that a **15%** reduction in site capacity causes coverage losses comparable to a **25%** demand surge. This asymmetry underscores the need to co-plan charging deployment with distribution grid reinforcement (Das et al. 2020). Joint planning between transportation agencies and utilities should identify corridors where both demand and grid capacity are constrained and prioritise investment accordingly.

Fourth, planners should **apply differentiated strategies to address spatial equity.** The observed allocation pattern—many low-demand zones achieving $> 75\%$ coverage while a smaller number of high-demand zones receive $< 50\%$—reflects a rational economic response to the distance constraint but raises distributional equity concerns (Almaghrebi et al. 2024). We recommend market-driven deployment in high-density urban cores, targeted public subsidies or public-private partnerships for underserved suburban and rural areas, and explicit equity metrics in planning including minimum coverage requirements for disadvantaged communities. The minimal impact of urban land cost premiums (Scenario S3: 50% higher costs yield only a 0.6 percentage point coverage reduction) suggests that cost differentials should not deter urban deployment where demand concentration justifies premium site acquisition.

In practice, the model outputs serve as a direct input to infrastructure planning decisions. The ranked site list in Table 5 provides an immediate priority order for site acquisition, with the highest-load sites offering the greatest coverage return per dollar invested. The zone-level coverage gaps in Table 6 identify where candidate network expansion is more effective than adding capacity at existing sites. The robustness budget Γ translates directly into procurement strategy, where $\Gamma = 30\%$ can be interpreted as protecting against simultaneous above-nominal demand in approximately one third of all zones, a plausible condition during peak travel periods or rapid fleet electrification campaigns. Finally, the minimal coverage impact of urban land cost premiums in Scenario S3 suggests that cost differentials alone should not deter deployment in high-demand urban zones.

Limitations and Future Research

Limitations

Several limitations of the present study should be noted.

The model is static and single-stage, so the gradual evolution of EV adoption and investment timing is not captured. While the sensitivity analysis in Section 4.5.4 shows that over 90% of opened sites remain consistent across 500 perturbed instances, a multi-period formulation in which site selection is updated as adoption data accumulates would more accurately reflect real infrastructure rollout. Demand is zone-aggregated without temporal heterogeneity, limiting analysis of peak loading, and queuing, congestion, and time-varying pricing are not modelled explicitly.

The objective is purely economic, with equity, grid impacts, and user experience reflected only indirectly. Zone-level aggregation of existing sites may overestimate

available capacity where individual plugs are heavily congested. The upper bound $K_{\max} = 3$ reflects the median site size in the study region but may underestimate feasible capacity in high-density urban zones where structured parking or dedicated charging facilities could accommodate larger installations. Relaxing this parameter for urban candidate zones is a straightforward extension for future work.

The case study is specific to Level 2 public charging, though the model is directly applicable to other charging types by recalibrating $D_{\max}$ and the demand conversion factor to reflect different user travel tolerances and session durations.

Finally, the present formulation solves tractably at 482 candidate sites but would require decomposition-based approaches or matheuristics at national scale with 10,000 or more candidate sites.

Future Research Directions

A natural extension is to develop multi-period or two-stage versions in which site acquisition is decided early and capacity is expanded incrementally as EV adoption grows (Pourgholamali et al. 2023; Cheng et al. 2021).

Future work should also introduce explicit equity constraints such as minimum coverage requirements for disadvantaged zones. These zones could be identified through census income or EV adoption gap data, ensuring the optimisation actively counteracts spatial inequities documented in the literature (Hsu and Fingerman 2021; Almaghrebi et al. 2024). Pareto frontier analysis across cost, grid loading, and coverage objectives would complement this direction (Almaghrebi et al. 2024).

A third avenue is closer integration with machine learning demand forecasting using smart-meter data and travel surveys. This would construct richer uncertainty sets and incorporate temporal variability and diurnal load peaks that the current zone-averaged daily demand representation does not capture (Ma and Fang 2022).

Coupling the strategic siting model with operational models incorporating queuing, time-of-use pricing, and vehicle-to-grid operation would provide a more complete view of system performance under high EV penetration (Li et al. 2025). The current formulation solves tractably at 482 candidate sites using the HiGHS solver within a 30-minute time limit. Scaling to a national level with 10,000 or more candidate sites would likely exceed practical time limits for exact MILP solvers. In such settings, decomposition-based approaches such as Lagrangian relaxation or Benders decomposition would be required. Metaheuristics that solve regional subproblems independently offer a complementary strategy for maintaining solution quality within acceptable computation times.

AI Assistance AI-assisted tools were used for grammar and writing improvement only, consistent with the journal's definition of AI-assisted copy editing. All scientific content, analysis, and conclusions are the original work of the authors.

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Data Availability EV demand data used in this study are derived from publicly available EV registration records accessible via regional transportation authorities. Existing station data are available from the U.S. Alternative Fuels Data Center (https://afdc.energy.gov). The Python/PuLP optimisation code is available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Ethics Approval This study does not involve human participants, animal subjects, or personal data, and therefore does not require ethics committee approval.

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